THE PRIME MINISTER'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
LONGIDO DISTRICT MID TERM JOINT EXAMINATION

FORM THREE
033 BIOLOGY

Time: 3 Hours **Wednesday, 26th August 2026 p.m.**

Instructions

1. This paper consists of Sections **A, B** and **C** with a total of eleven questions.
2. Answer **all** questions in Sections **A** and **B** and **two** questions from Section **C.**
3. Section A carries **16 marks**, Section B carries **54 marks**, and Section C carries **30 marks**.
4. All answers should be written in blue or black ink.
5. All drawings should be done in pencil.
6. Write your examination number on every page of your answer booklet.

SECTION A (16 MARKS)

Answer **all** questions in this section.

1. For each of the items (i)-(x), choose the most correct answer among the given alternatives and write its letter beside the item number.

(i) During a biology practical, a student wants to investigate whether light is necessary for photosynthesis. Which action represents the **hypothesis** in the investigation?
   A. Testing a leaf with iodine solution
   B. Covering part of a leaf with black paper
   C. Light is necessary for photosynthesis
   D. Placing the plant in sunlight
   E. Recording the colour of the leaf after testing

(ii) A student accidentally spills a chemical on her skin during a practical lesson. What should she do first?
   A. Wipe the chemical with a dry cloth
   B. Wash the affected area with plenty of clean water
   C. Continue with the experiment
   D. Cover the affected area with paper
   E. Apply laboratory chemicals to neutralise it

(iii) A patient with severe blood loss requires an urgent blood transfusion. Which blood component is mainly responsible for transporting oxygen to body tissues?
   A. Plasma
   B. Platelets
   C. White blood cells
   D. Red blood cell
   E. Lymph

(iv) A farmer notices that a plant placed in a dark cupboard has pale yellow leaves. Which process is most directly affected by the absence of light?

A. Respiration
B. Photosynthesis
C. Transpiration
D. Absorption
E. Excretion

(v) A student places a plant cell in a concentrated salt solution. Water moves out of the cell through the selectively permeable membrane. This process is called:
A. Diffusion
B. Active transport
C. Osmosis
D. Translocation
E. Mass flow

(vi) Which structure in the human respiratory system is the main site of gaseous exchange?
A. Trachea
B. Bronchus
C. Alveolus
D. Larynx
E. Diaphragm

(vii) A student eats bread and later glucose is absorbed into the bloodstream. Which structure is mainly responsible for the absorption of digested nutrients?
A. Stomach
B. Large intestine
C. Small intestine
D. Oesophagus
E. Rectum

(viii) Which vessel carries oxygenated blood from the lungs towards the heart?
A. Pulmonary artery
B. Pulmonary vein
C. Aorta
D. Vena cava
E. Coronary artery

(ix) A person touches a hot object and immediately withdraws the hand. Which part of the nervous system coordinates this rapid response?
A. Reflex arc
B. Cerebellum
C. Cerebrum
D. Spinal cord only
E. Sense organ only

(x) Which organ is primarily responsible for removing urea from the blood?
A. Liver
B. Heart
C. Kidney
D. Pancreas
E. Lung

2. Match each item in List **A** with the most appropriate response in List **B.**

LIST A	LIST B
(i) Movement of particles from a region of high concentration to a region of low concentration. (ii) Movement of water molecules through a selectively permeable membrane. (iii) Tissue that transports manufactured food in plants. (iv) Organ that pumps blood around the body. (v) Small air sacs where gaseous exchange takes place. (vi) Functional unit of the kidney.	A. Alveolus B. Diffusion C. Nephron D. Phloem E. Osmosis F. Xylem G. Heart H. Stoma

SECTION B (54 MARKS)

Answer **all** questions in this section.

3. During a school biology investigation, students decide to determine the effect of temperature on the activity of an enzyme. They prepare several test tubes, make observations and record their results.
   (a) State four steps that the students should follow when conducting a scientific investigation and three laboratory safety precautions that should be observed during the investigation.
   (b) Give two reasons why accurate recording of observations is important in a biological investigation.

4. A school health club discovers that several students frequently fail to wash their hands before eating and that waste is sometimes thrown near the school water source.
   (a) Explain three ways in which poor personal hygiene can increase the spread of infectious diseases.
   (b) State three ways by which the body protects itself against disease-causing organisms.

5. During a practical demonstration, a student accidentally touches a hot metal object and immediately withdraws his hand.
   (a) Describe the pathway followed by the nerve impulse from the receptor until the hand is withdrawn.
   (b) Distinguish between the following pairs:
   (i) sensory neurone and motor neurone;
   (ii) voluntary movement and involuntary movement.

6. (a) Seypalm eats a meal containing starch, proteins, fats, vitamins and mineral salts. State the end products of digestion of of the food substance in Seypalm's meal:
   (b) Explain six adaptations of the small intestine for absorption of digested food.

7. (a) A patient is admitted to hospital with severe anaemia. The doctor explains that the condition affects the transport of oxygen around the body. Explain three adaptations of red blood cells for oxygen transport.
   (b) State four differences between arteries and veins.

8. (a) Define the term **excretion** and distinguish it from egestion.
   (b) Describe the main stages involved in the formation of urine in a nephron.

SECTION C (30 MARKS)

Answer any **two questions** from this section.

9. Imagine you are a botanist invited to give a presentation to local farmers about the differences between conifers found in the mountain forests and Angiosperms found in their agricultural valleys. Write an essay that covers the following:
   (a) Three distinct characteristics of division Coniferophyta and three distinct characteristics of division Angiospermophyta.
   (b) Three anatomical differences between Conifers and Angiosperms.
   (c) Explain three ways in which Angiosperms are economically important to human modern society.

10. Survival utilize specialized structures such as the human skeleton.
   (a) Explain three biological reasons why locomotion is crucial for the survival of animals.
   (b) The human skeleton is divided into two main parts. Name these two parts and state two bones found in each part.
   (c) Discuss three functions of the human skeleton other than providing support and aiding movement.

11. Human activities have significantly disrupted the natural ecosystems, threatening the global balance of nature through pollution and habitat destruction.
   (a) Define the term Ecosystem and state two biotic factors and two abiotic factors that interact within it.
   (b) Explain how three human actions lead to the destruction of the balance of nature in an area.
   (c) Propose two sustainable methods that communities can implement to conserve ecosystems and restore environmental balance.